Painting the complete picture

Citizen scientists are now the biggest global providers of data on the occurrence of living organisms. Data is easier than ever to submit and artificial intelligence (AI) algorithms can process data in minutes where once it would have taken weeks. For example, if a person records sightings of birds coming to a garden feeder and sends a report from a phone to Cornell University's eBird website, the information is compared with previous data on factors such as population numbers and migration routes. By 2021, more than 680,000 people had submitted over 1 billion bird observations to eBird from 202 countries around the world. This data is fed into the Global Biodiversity Information Facility (GBIF, coordinated in Denmark), which collects information on plants, animals, fungi, and bacteria. GBIF now contains more than 3 billion observations, and the number is growing daily.

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